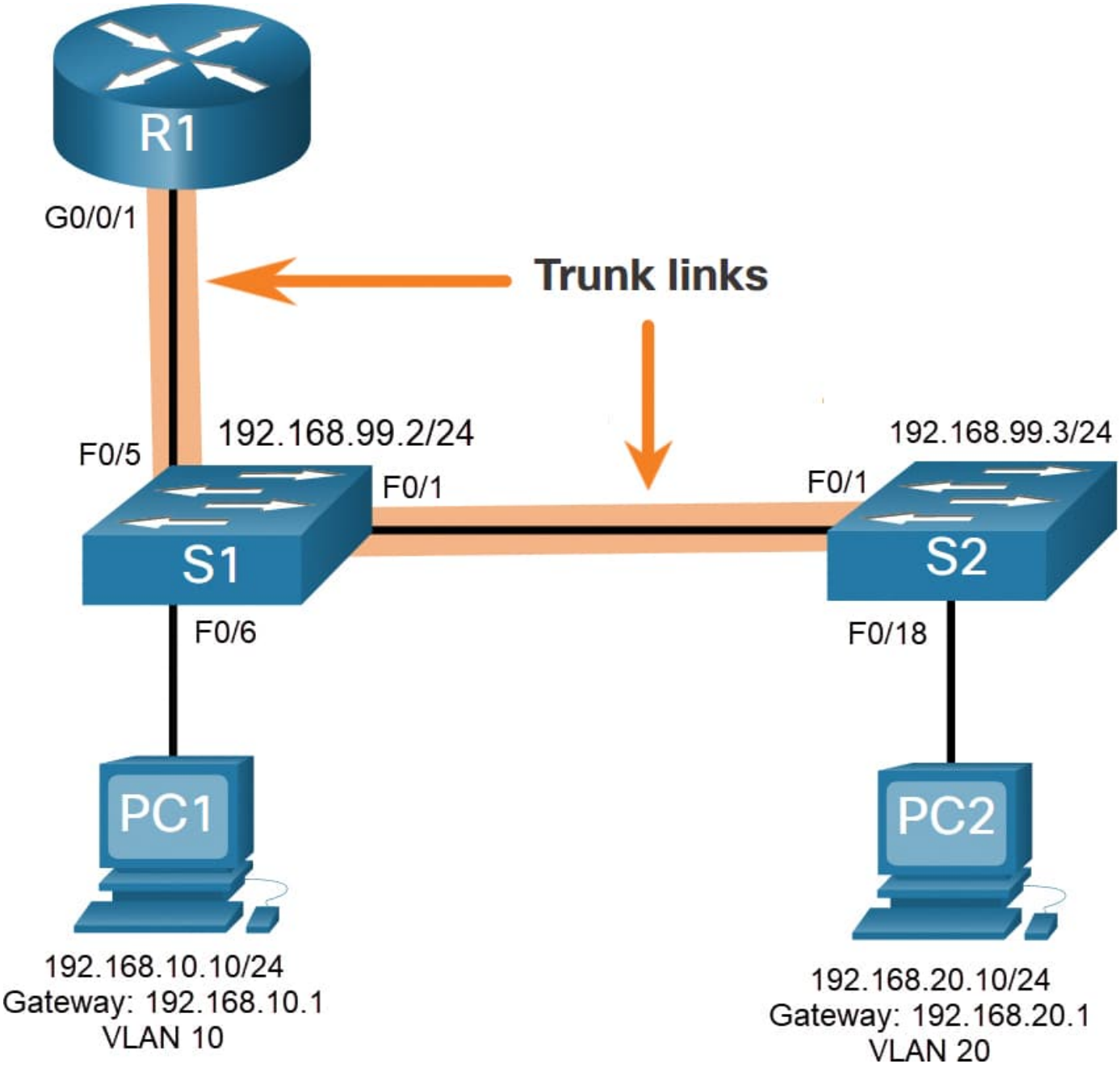
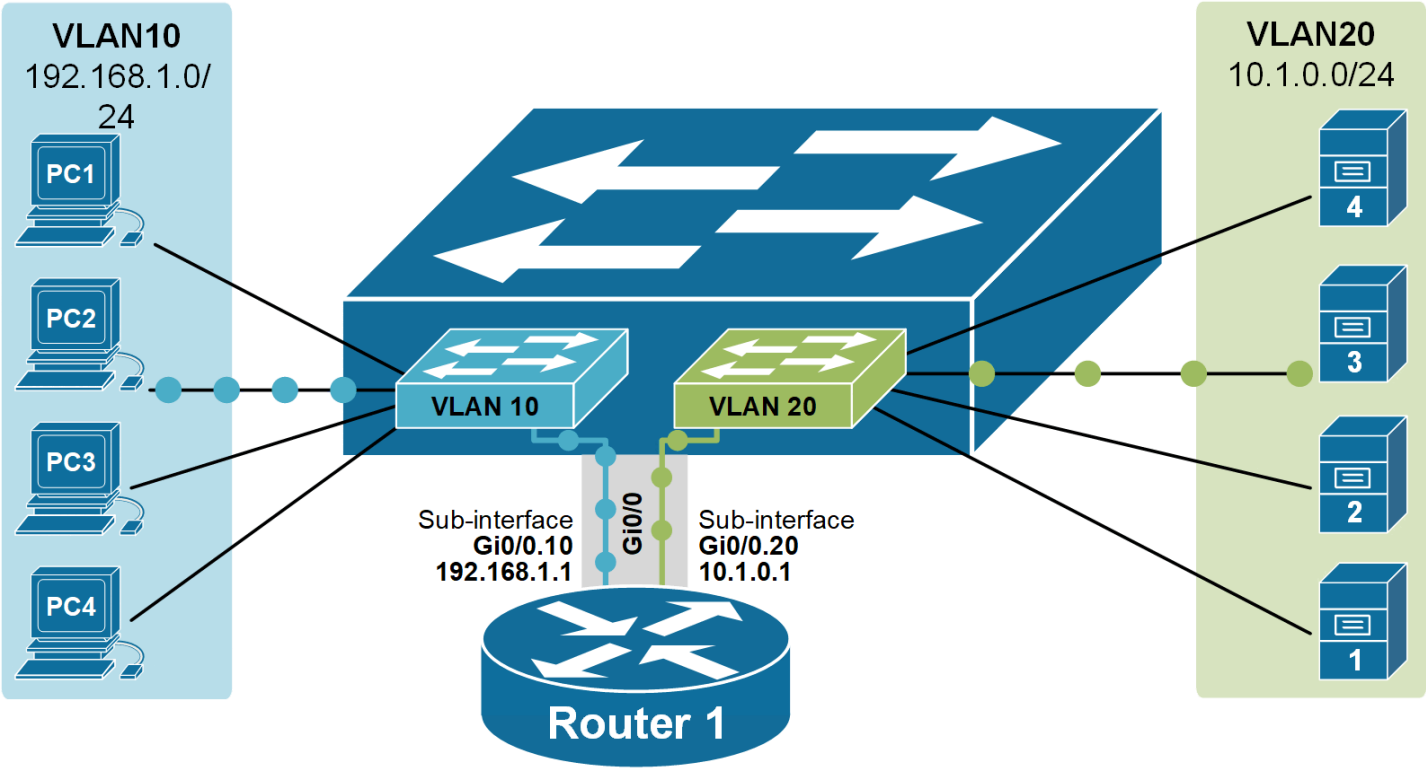
Inter-VLAN (Communication Between VLANs)





1. What is Inter-VLAN?

Inter-VLAN routing means **communication between different VLANs using a router or Layer-3 switch**.

Normally:

- Devices in **same VLAN → can communicate**
- Devices in **different VLAN → cannot communicate**

To allow communication between VLANs we use **Inter-VLAN routing**.

Example

Device	VLAN
PC1	VLAN 10
PC2	VLAN 20

Without routing:
❌ PC1 cannot communicate with PC2.

With **Inter-VLAN routing**:
✔ PC1 can communicate with PC2.

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2. Why Inter-VLAN Routing is Required

VLANs create **separate broadcast domains**.

Example company network:

Department	VLAN
HR	VLAN 10
IT	VLAN 20
Finance	VLAN 30

But departments still need to communicate sometimes.
So we use **Inter-VLAN routing**.

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3. Methods of Inter-VLAN Routing

1️⃣ Legacy Inter-VLAN Routing

- Each VLAN connects to a **separate router interface**.
- Not efficient because many router ports are required.

Example:

```
Router
 |    |    |
V10  V20  V30
```

2️⃣ Router-on-a-Stick (Most Common)

One **router interface** is used with **multiple subinterfaces**.

A **trunk link** connects the switch and router.

Example topology:

```
PC1 ---\
        Switch ---- Router
```

```
PC2 ---/
```

Switch port → **trunk**
Router → **subinterfaces**

3️⃣ Layer-3 Switch Routing

A **Layer-3 switch** performs routing between VLANs directly.

Advantages:

- Faster
- No external router required

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4. Router-on-a-Stick Configuration Example

Step 1 – Create VLANs on Switch

```
Switch(config)# vlan 10
Switch(config)# vlan 20
```

Step 2 – Assign Ports

```
Switch(config)# interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config)# interface fa0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
```

Step 3 – Configure Trunk Port

```
Switch(config)# interface fa0/24
Switch(config-if)# switchport mode trunk
```

Step 4 – Router Configuration

```
Router(config)# interface fa0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0

Router(config)# interface fa0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

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5. Example Network

Device	IP	VLAN
PC1	192.168.10.2	10
PC2	192.168.20.2	20

Gateway:

VLAN	Gateway
VLAN 10	192.168.10.1
VLAN 20	192.168.20.1

Now PC1 can **ping PC2**.

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6. Important Exam Point

Definition

> Inter-VLAN routing is the process of forwarding network traffic between different VLANs using a router or Layer-3 switch.